When can machine learning beat a fixed rule?

Simulation with known truth and a pre-registered test on JPMaQS duration, FX and equity returns · Di Liu · November 2026

Summary.

- **Question.** In the example-code study, learned signals did not beat a simple theory-signed composite. Is that because machine learning cannot help with macro signals, or because the conditions it needs were absent?
- **Part A: simulation.** On panels where the true relationship is known, learners beat the rule only when the rule is mis-specified in a way a constrained model can represent: unequal weights, or signs that flip by country or regime. Even then they need about 24 markets × 25 years at a realistic signal strength, and the edge is about **+0.02 in out-of-sample IC**. Gradient boosting is the weakest learner throughout.
- **Part B: real data.** Four learners against theory-signed rules in duration (24 markets), FX (23) and equity (16), 2007–2023, under criteria written before the run. **None of 16 comparisons passes** (largest t = 1.51 against a threshold of 2.96).
- **Two traps.** In FX the learners lead only because the rule loses money; in equity their higher Sharpe is mostly long-equity beta. On ICs and beta-neutral Sharpe the rule is best in both.

1. Design

The two parts answer complementary questions. Part A asks *when should a learner win?* with the truth known; Part B asks *does it win on real data?* with the analysis fixed in advance, so that the answer cannot be tuned after seeing results.

Contenders (both parts).

Name	What it does	What it assumes
Rule	equal-weight average of theory-signed z-scores	signs from theory, equal weights, linear
Sign-constrained	non-negative least squares (or non-negative Ridge) on the signed features	signs from theory; learns the weights
Ridge (free signs)	Ridge on all features	nothing about signs
Linear + interactions	Ridge on features plus feature × context terms	is told *which* interactions may exist
Gradient boosting	shallow boosted trees on features plus context	must discover thresholds and interactions itself

Part A panels. N markets × T months of seven persistent macro features (AR(1), φ = 0.9, with a common factor). Returns are a true signal plus noise, set so the best attainable monthly R^2 is 0.5%, 2% or 8%. Six truths, from the rule's home ground to cases built for learning: rule exactly right; thresholds ($|x| > 1$); unequal weights with irrelevant inputs (7 or 30 features); opposite signs in half the markets (observable country type); and a sign that reverses with an observable regime. Four sizes: 4 × 180, 10 × 240, 24 × 300 and 48 × 480. Every learner is refitted every 12 months on an expanding window; six simulated panels per cell (432 runs).

Part B data and test. Kaggle JPMaQS (point-in-time, 2000–2023). Nine identical features for every asset class: inflation and growth relative to benchmarks, credit growth, real 5y and 2y yields, the real slope and FX carry. Theory signs differ by asset (8 used for duration, 3 for FX, 5 for equity). Learners are re-selected quarterly by `SignalOptimizer` with expanding panel cross-validation. A learner **beats the rule** only if its paired HAC t-statistic of daily PnL differences is ≥ 2.96 (Bonferroni for 16 tests) **and** its PnL gap is positive in both developed and emerging sub-panels.

2. Part A: where each learner wins

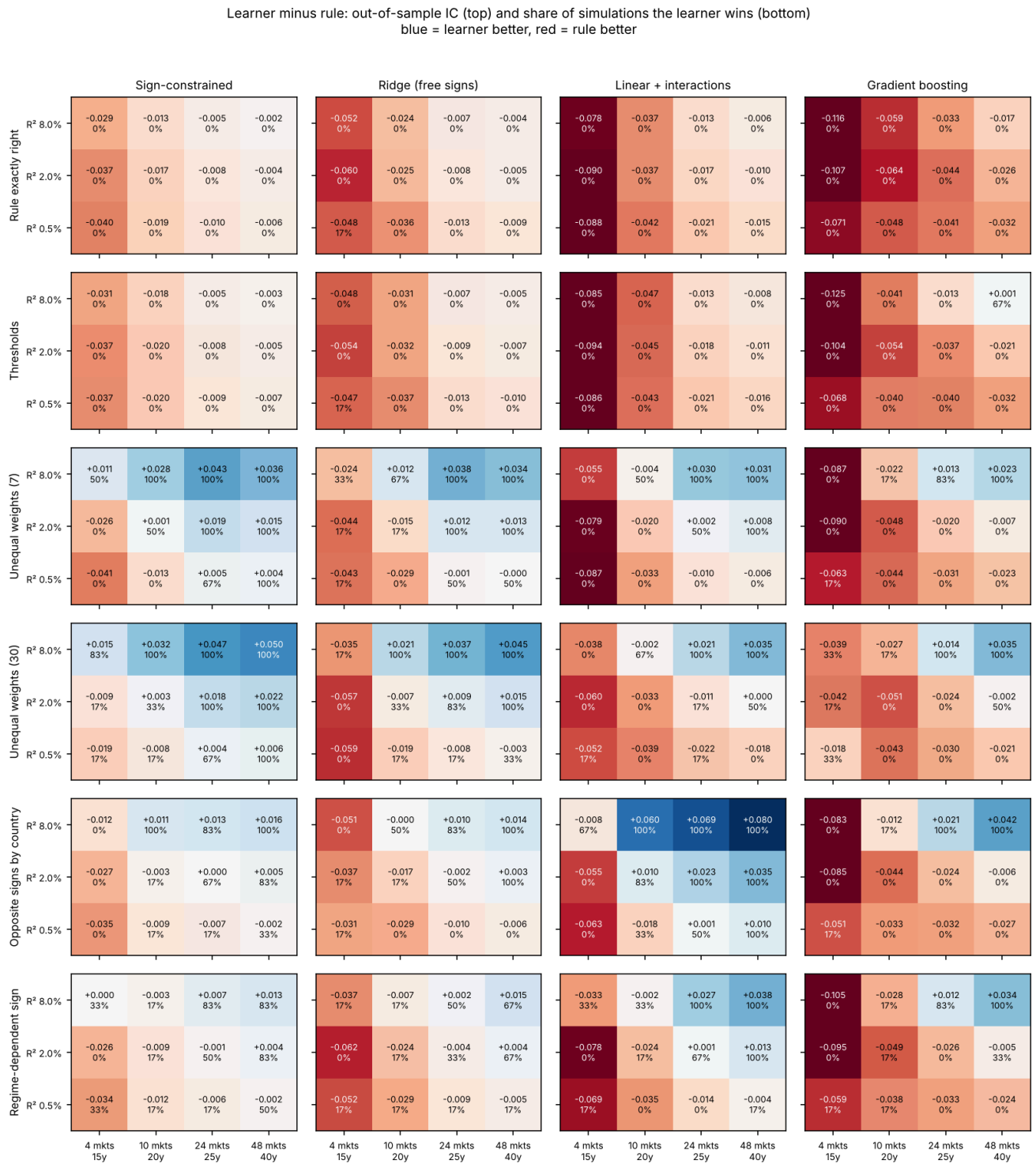


Figure 1. Learner minus rule out-of-sample IC (top number in each cell) and the share of simulated panels in which the learner wins (bottom). Blue = learner better. The real JPMaQS panels sit near the 24-market, R^2 2% column.

Truth	Result
Rule exactly right	No learner wins on average in any cell. A learner can at best rediscover equal weights, with noise.
Thresholds	The linear rule captures most of the effect (IC 0.13 against an oracle 0.15 at 24 markets, R^2 2%). Boosting draws level only at 48 markets, R^2 8%.
Unequal weights	First place learning pays: the sign-constrained learner wins 100% of panels from 24 markets at R^2 2% (+0.019 IC with 7 inputs, +0.018 with 30).
Opposite signs by country	Linear + interactions wins 100% from 24 markets at R^2 2% (+0.023), +0.06–0.08 at R^2 8%. Boosting still loses at R^2 2% even at 48 × 480.
Regime-dependent sign	Same pattern, weaker: interactions win from 24 markets at R^2 8%, or 48 at R^2 2%.
Any truth, 4 markets	The rule wins almost everywhere, matching the example-code result.

Prediction for Part B. At real-data scale a learner can beat a theory-signed rule only if the rule mis-weights its inputs or misses sign heterogeneity, and then by about 0.02 in IC: small relative to sampling noise over 17 years.

3. Part B: real data under pre-registered criteria

	Rule	Sign-constrained	Ridge	Linear + int.	Boosting
Duration IC	**0.126**	0.102	0.083	0.067	0.033
Sharpe / beta-neutral	**1.18 / 1.21**	1.17 / 1.13	0.94 / 0.85	0.77 / 0.69	0.47 / 0.32
t vs rule		−0.03	−0.68	−1.19	−1.86
FX IC	−0.003	0.008	0.001	0.013	0.012
Sharpe / beta-neutral	−0.17 / −0.17	**0.14 / 0.16**	0.05 / 0.08	0.00 / 0.01	0.09 / 0.13
t vs rule		1.51	0.89	0.77	0.99
Equity IC	**0.085**	0.061	0.047	0.046	0.038
Sharpe / beta-neutral	0.42 / **0.53**	0.47 / 0.47	**0.67** / 0.37	0.62 / 0.33	0.58 / 0.26
Correlation with long-only	−0.18	−0.01	0.71	0.64	0.71
t vs rule		0.33	0.73	0.54	0.45
Multi-asset Sharpe	0.82	**0.99**	0.95	0.79	0.66
t vs rule		0.76	0.46	−0.06	−0.46

Table 1. IC = pooled monthly Pearson correlation with next-month returns. Naive PnLs at 10% vol, monthly rebalancing, one-day slippage, gross of costs, January 2007 – December 2023. Beta-neutral Sharpe removes the PnL's regression on the long-only return. All duration and equity ICs are significant at 5%; no FX IC is.

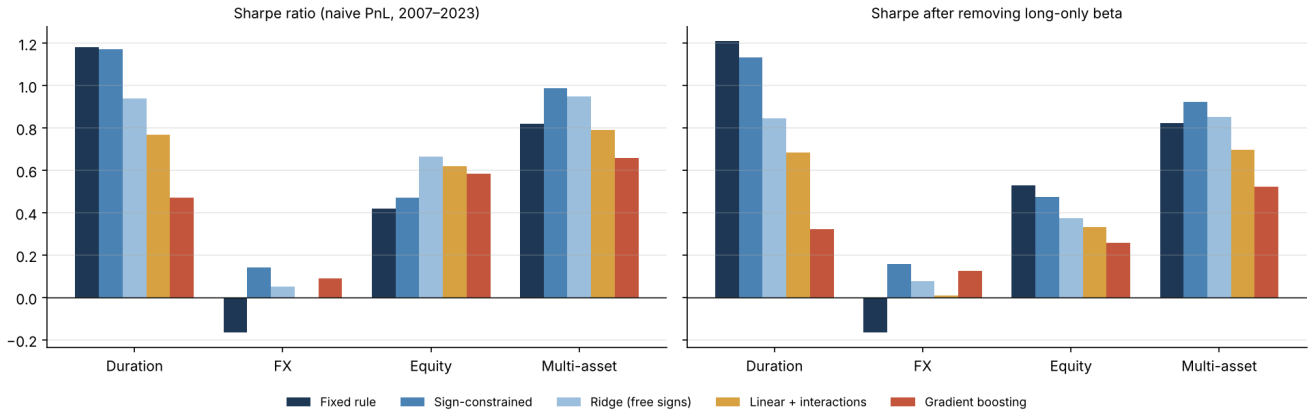


Figure 2. Headline Sharpe (left) and Sharpe after removing long-only beta (right).

Duration (theory clear): the rule wins. Highest IC and Sharpe. The sign-constrained learner ties it; every learner free to choose signs or discover interactions is worse, boosting much worse. This is Part A's "rule exactly right" regime.

FX (theory unclear): nothing predicts. The rule's IC is −0.003 and the learners' 0.001–0.013, none significant. The learners lead by about 3% a year only because the rule loses money; their gaps are positive in developed markets but mixed in emerging ones. Learning cannot extract a signal the features do not contain.

Equity: the learners' extra Sharpe is beta. Their PnLs correlate 0.64–0.71 with simply holding equities: they learned to stay long through a strong equity decade. Their ICs are all below the rule's, and after removing beta the rule is best (0.53 against 0.26–0.47).

Against expectations written before the run. Duration: confirmed. FX: the learners came closest, as expected, but because the rule failed rather than because they found a signal. Equity: the rule was more predictive than expected. No learner beats a rule: confirmed.

Disclosure. The first run left USD out of the three learners that use FX carry (USD has no carry series, and its zero z-score was dropped). USD carry was set to zero by definition and everything re-run before results were written up. The beta-neutral Sharpe was added afterwards as a diagnostic; the decision rule was not changed.

4. Conclusions

1. **Whether ML helps depends on how wrong the rule is, and that can be measured.** With known truth, learning wins only against a mis-specified rule, and only if the learner can represent the error.
2. **Data volume is the binding constraint.** Even favourable cases need about 24 markets × 25 years for a +0.02 IC edge. Seventeen real years cannot confirm an edge that size, so Part B's null is what Part A predicts.
3. **Structure beats flexibility.** The learners that won were the most constrained: theory signs with learned weights, or a specified interaction. Gradient boosting was weakest in both parts.
4. **Judge learners on IC, sub-panels and beta-neutral returns, not headline Sharpe.** Both apparent real-data "wins" disappear on those measures.

Practical rule. Start with a theory-signed composite. Use a learner to re-weight inputs under sign constraints, or to test a specific, economically motivated interaction; promote it only if it passes a pre-registered test.

Limits. Gross of costs; one feature set per asset class; Gaussian AR(1) simulated features without structural breaks; six simulated panels per cell; Bonferroni makes the real-data test conservative.

Code and full tables: `ml_vs_rules.ipynb` (same folder). Data: JPMaQS Kaggle release, Macrosynergy. Packages: macrosynergy 1.8.5, scikit-learn, statsmodels.